Answer all questions, maximum marks: 5 x 5 = 25). Time is two hrs.

1. (a) The following is the definition of the `Softmax` function with temperature parameter (τ) which can typically take a positive value:

$$\text{softmax}(x_i) = \frac{e^{x_i/\tau}}{\sum_j e^{x_j/\tau}} \tag{1}$$

What will be the effect of increasing and decreasing the temperature parameter τ? **(2 marks)**

Ans. Higher temperature $\rightarrow$ smoothing effect, Lower temperature $\rightarrow$ sharpening effect.

(b) What is the importance of using an auxiliary classifier in the Inception-v3 network? What is Label smoothing and how does it help? Explain mathematically. **(3=1+2 marks)**

Ans. Reduces vanishing gradient phenomena in deep network.

Given a classification problem with C classes, and a smoothing parameter ϵ, label smoothing adjusts the target labels as follows. For a true class c and any class i, the adjusted target label y'_i is defined by:

$$y'_i = \begin{cases} 1 - \epsilon + \frac{\epsilon}{C} & \text{if } i = c, \\ \frac{\epsilon}{C} & \text{otherwise.} \end{cases}$$

Here, ϵ is a small positive number less than 1, representing the smoothing factor, and C is the total number of classes.

2. (a) Consider ISRO has recently launched a satellite and you are assigned a work to do Land Use and Land Cover(LULC) classification problem from the captured satellite images. You first train your network on 40 satellite images. Training converges, but the training loss is very high. You then decide to train this network on 10,000 examples from other satellite images of same resolution. Is your approach to fixing the problem correct? If yes, explain the most likely results of training with 10,000 examples. If not, give a solution to this problem. **(3 marks)**

Ans: The model is suffering from a bias problem. Increasing the amount of data reduces the variance, and is not likely to solve the problem. A better approach would be to decrease the bias of the model by maybe adding more layers/ learnable parameters. It is possible that training converged to a local optimum. Training longer/using a better optimizer/ restarting from a different initialization could also work.

(b) After visually inspecting the dataset, you realize that the training set only contains pictures taken during the day, whereas the validation/test set only has pictures taken at night. Explain what is the issue and how would you correct it? **(2 marks)**

Ans: It can cause a domain mismatch. The difference in the distribution of the images between training and dev might lead to faulty hyperparameter tuning on the dev set, resulting in poor performance on unseen data.

Solution: randomly mix pictures taken at day and at night in the two sets and then resplit the data.

3. (a) Derive the equations for the backward pass of the backpropagation algorithm for a network that uses leaky ReLU activations, which are defined as

$$a[z] = \text{ReLU}[z] = \begin{cases} \alpha \cdot z & \text{if } z < 0 \\ z & \text{if } z \geq 0 \end{cases} \tag{2}$$

(2 marks)

Given the leaky ReLU activation function defined as:

$$a[z] = \text{ReLU}[z] = \begin{cases} \alpha \cdot z & \text{if } z < 0, \\ z & \text{if } z \geq 0, \end{cases} \tag{3}$$

where α is a small constant.

The derivative of the leaky ReLU function with respect to its input z is:

$$\frac{da}{dz} = \begin{cases} \alpha & \text{if } z < 0, \\ 1 & \text{if } z \geq 0, \end{cases} \tag{4}$$

During the backward pass in backpropagation, for any layer l in a network with L layers, the gradient of the loss J with respect to the linear combination of inputs $z^{[l]}$ is denoted as $\delta^{[l]}$. For the final layer L, this gradient depends on the specific loss function used and is given by $\delta^{[L]}$. For any hidden layer l, the gradient $\delta^{[l]}$ is computed as:

$$\delta^{[l]} = ((W^{[l+1]})^T \delta^{[l+1]}) * \frac{da^{[l]}}{dz^{[l]}}, \tag{5}$$

where $*$ denotes element-wise multiplication and the derivative $\frac{da^{[l]}}{dz^{[l]}}$ is applied as per the leaky ReLU derivative defined above.

The gradients with respect to the weights $W^{[l]}$ and biases $b^{[l]}$ at layer l are calculated as:

$$\frac{\partial J}{\partial W^{[l]}} = \delta^{[l]} (a^{[l-1]})^T, \tag{6}$$

$$\frac{\partial J}{\partial b^{[l]}} = \delta^{[l]}, \tag{7}$$

where $\frac{\partial J}{\partial W^{[l]}}$ and $\frac{\partial J}{\partial b^{[l]}}$ are the gradients of the loss function with respect to the weights and biases of layer l, respectively.

(b) Consider two hidden layers of size 224×224 with C^1 and C^2 channels, respectively, connected by a 3×3 convolutional layer. Describe how to initialize the weights using Xavier initialization. **(1.5 marks)**

(c) Show that the maximum number of regions created by a shallow network with $D_i = 2$ dimensional input, $D_o = 1$ dimensional output, and $D = 3$ hidden units is seven. **(1.5 marks)**

Ans. Consider a shallow neural network with a two-dimensional input ($D_i = 2$), one-dimensional output ($D_o = 1$), and three hidden units ($D = 3$). We aim to show that such a network can create a maximum of seven regions in the input space.

Approach

A shallow neural network has a single hidden layer. Each unit in the hidden layer creates a decision boundary in the input space. For a two-dimensional input, each decision boundary is a line that divides the input plane into two regions. The intersections of these boundaries create distinct regions.

Counting the Regions

(a) The **first hidden unit** creates a line, dividing the plane into two regions.

(b) The **second hidden unit** adds another line. This line can create a maximum of two additional regions by intersecting with or lying apart from the first line, totaling four regions.

(c) The **third hidden unit** introduces a third line. To maximize the number of regions:

- The third line intersects with the first line, adding two new regions.
- It then intersects with the second line, adding one more region.

Thus, before the third line, we have four regions. The addition of the third line, intersecting with the existing lines, brings in two more regions at the first intersection and one additional region at the second intersection.

Total Regions

The total maximum number of regions that can be created by these three lines – and thus by the network with three hidden units – is:

$$4 + 2 + 1 = 7.$$

This demonstrates that a shallow neural network with the given specifications can partition the input space into a maximum of seven distinct regions.

4. Consider a model where the prior distribution over the parameters is a normal distribution with mean zero and variance σ_ϕ^2 so that

$$\Pr(\phi) = \prod_{j=1}^{J} \text{Norm}_{\phi_j}(0, \sigma_\phi^2).$$

where j indexes the model parameters. When we apply a prior, we maximize

$$\prod_{i=1}^{I} \Pr(y_i|x_i, \phi) \cdot \Pr(\phi).$$

Show that the associated loss function of this model is equivalent to that of an L_2 regularized system. **(5 marks)**

Ans. Consider a model where the prior distribution over the parameters, denoted as ϕ, is a normal distribution with mean zero and variance σ_ϕ^2:

$$\Pr(\phi) = \prod_{j=1}^{J} \text{Norm}_{\phi_j}(0, \sigma_\phi^2),$$

where j indexes the model parameters, and J is the total number of parameters. This prior reflects our belief about the distribution of ϕ before observing any data.

When we apply this prior, our objective is to maximize the posterior probability, which involves the product of the likelihood of the observed data and the prior probability of the parameters:

$$\prod_{i=1}^{I} \Pr(y_i|x_i, \phi) \cdot \Pr(\phi).$$

To simplify, we consider the log of this expression, transforming the product into a sum:

$$\sum_{i=1}^{I} \log \Pr(y_i|x_i, \phi) + \sum_{j=1}^{J} \log \mathrm{Norm}_{\phi_j}(0, \sigma_\phi^2).$$

The second term, involving the log of the normal distribution, is particularly interesting:

$$\log \mathrm{Norm}_{\phi_j}(0, \sigma_\phi^2) = -\frac{1}{2\sigma_\phi^2}\phi_j^2 + \text{const},$$

where the constant term is independent of ϕ_j and thus does not affect the optimization.

Summing over all parameters, we get the regularization term of our loss function:

$$-\frac{1}{2\sigma_\phi^2}\sum_{j=1}^{J}\phi_j^2,$$

which is an L_2 regularization term scaled by $\frac{1}{2\sigma_\phi^2}$.

Therefore, maximizing the posterior probability with a normal prior on the parameters is equivalent to minimizing a loss function that includes both a data fidelity term (negative log-likelihood of the data given the parameters) and an L_2 regularization term on the parameters. This demonstrates the equivalence of using a normal prior distribution over the parameters to applying L_2 regularization in the model.

5. Two historians approach you for your deep learning expertise. They want to classify images of historical objects into 3 classes depending on the time they were created:

- Antiquity (y = 0)
- Middle Ages (y = 1)
- Modern Era (y = 2)

(A) Class: Antiquity (B) Class: Middle Ages (C) Class: Modern Era

Figure 1: Example of images found in the dataset along with their classes

Over the last few years, the historians have collected nearly 5,000 hand-labelled RGB images.

- Before training your model, you want to decide the image resolution to be used. Why is the choice of image resolution important? **(1 mark)**

  Ans. Trade-off between accuracy and model complexity.

- If you have to use a BoVW based hard-crafted features, will there be any potential problem in the obtained features? **(1 mark)**

  Ans. Polysemy, redundancy, synonyms. lack of generalizability

- You come up with a CNN classifier. For each layer, calculate the number of weights, number of biases and the size of the associated feature maps.**(3 marks)**

  The notation follows the convention:

- CONV-K-N denotes a convolutional layer with N filters, each them of size K×K, Padding and stride parameters are always 0 and 1 respectively.
- POOL-K indicates a K × K pooling layer with stride K and padding 0.
- FC-N stands for a fully-connected layer with N neurons.

Layer	**Activation map dimensions**	**Number of weights**	**Number of biases**
INPUT	$128 \times 128 \times 3$	0	0
CONV-9-32			
POOL-2			
CONV-5-64			
POOL-2			
CONV-5-64			
POOL-2			
FC-3			

Ans: Successively:

- $120 \times 120 \times 32$ and $32 \times (9 \times 9 \times 3 + 1)$
- $60 \times 60 \times 32$ and 0
- $56 \times 56 \times 64$ and $64 \times (5 \times 5 \times 32 + 1)$
- $28 \times 28 \times 64$ and 0
- $24 \times 24 \times 64$ and $64 \times (5 \times 5 \times 64 + 1)$
- $12 \times 12 \times 64$ and 0
- 3 and $3 \times (12 \times 12 \times 64 + 1)$

The End.